

ISKANDAR KHOLMANOV

Applied Scientist / Machine Learning Engineer

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Green card | Authorized to work in the U.S. | No visa sponsorship required | Open to relocation

SUMMARY

ML Engineer with a PhD in Physics, 8+ years of scientific research experience, and 6+ years of business ownership solving complex analytical problems. Experienced building and deploying production AI systems including recommendation systems, RAG pipelines, semantic search, neural ranking models, and real-time ML inference services. Hands-on expertise across the full ML lifecycle - from data engineering to evaluation, deployment, observability, and optimization. Proven ability to translate ambiguous business problems into scalable ML solutions. Co-author of 40+ scientific publications and holder of a U.S. patent..

SKILLS

- ❖ **ML & DL:** Supervised/unsupervised learning, deep learning, transformers, recommendation systems, retrieval & ranking models, embeddings, PyTorch, TensorFlow, scikit-learn, XGBoost, experimental design, offline evaluation;
- ❖ **Generative AI & LLMs:** RAG, semantic search, vector databases (FAISS, Qdrant, ChromaDB), LLM fine-tuning (LoRA/QLoRA), Hugging Face, LangChain, LlamaIndex, LLaMA-3.x, Groq APIs, LLM evaluation;
- ❖ **Engineering & MLOps:** Python, SQL, FastAPI, Docker, Kubernetes, MLflow, GitHub, CI/CD, model serving;
- ❖ **Cloud & Data:** AWS (EC2, S3, Lambda, API Gateway, CloudWatch), GCP (BigQuery, GCS), Databricks, Spark, scalable data pipelines;

EXPERIENCE

- ❖ **Data Scientist** | TripleTen | 2024–Present
 - Designed and deployed production AI systems using RAG, LLM inference APIs, FastAPI, and AWS, owning the full ML lifecycle from experimentation through deployment and production monitoring;
 - Built and evaluated LLM fine-tuning (LoRA/QLoRA), analyzing trade-offs among model quality, latency, cost;
 - Deployed supervised ML models, implementing production observability, performance metrics, and monitoring;
- ❖ **Founder & Business owner** | Uzeat LLC | 2017-2024
 - Led end-to-end business operations including forecasting, optimization, and data-driven strategic decisions;
 - Built analytical approaches to improve customer experience and profitability while balancing business constraints;
- ❖ **Research Scientist**, The University of Texas at Austin | 2010 – 2017
 - Led complex data-driven research and experimentation resulting in 40+ publications and a U.S. patent [Github](#);
 - Developed analytical and experimental methods to analyze complex systems and large datasets;

SELECTED PROJECTS

- ❖ **Personalized Recommendation System:** Built an end-to-end recommendation platform using **BigQuery**, **GCS**, and streaming data pipelines with a **Two-Tower retrieval model**, **Qdrant ANN vector search**, and **Deep & Cross Network** neural ranking model. Deployed online FastAPI inference, serving **real-time Top-10 personalized recommendations**. | [Github](#)
- ❖ **Production RAG System with Cloud Observability & LLM Serving:** Built a production-ready RAG system using **Sentence Transformers (MiniLM)**, **FAISS**, and **LLaMA-3.1-8B (Groq API)** with **FastAPI** on **AWS (EC2, S3)**. Implemented **CloudWatch** observability including structured logging, latency decomposition, token usage tracking, and p95 latency analysis to evaluate production performance and reliability. [Github](#)
- ❖ **LLM Fine-Tuning Pipeline:** Built an instruction-tuning pipeline using **Mistral-7B**, **PyTorch**, **HF Transformers**, **PEFT**, and **LoRA/QLoRA**. Created datasets from product reviews and synthetic Q&A, deployed a **4-bit quantized FastAPI inference service**, and evaluated instruction following, hallucinations, latency, and model quality. | [Github](#)

EDUCATION

- **TripleTen Bootcamp** | Data Science / Machine Learning Certificate | 2024;
- **PhD in Physics**, Academy of Science of Uzbekistan;